

# **Brazil E-Commerce Sales Analysis – Interview**

## **Explanation**

### **Project Overview**

This project was built to analyze Brazil e-commerce sales data and generate business insights using Excel, PostgreSQL, SQL, and Power BI. The goal of the project was to understand revenue trends, customer purchasing behavior, payment methods, seller performance, order delivery performance, and regional sales distribution.

### **The workflow included:**

- Data cleaning and preprocessing in Excel
- SQL-based analysis in PostgreSQL
- KPI calculations and aggregations
- Interactive dashboard creation in Power BI
- Business insight generation through visualizations

### **Business Problems Solved**

- Identified top-performing states generating maximum revenue
- Analysed monthly revenue growth trends
- Evaluated most preferred payment methods
- Tracked order delivery performance and average delivery time
- Identified top-performing sellers
- Analyzed order status distribution to understand operational efficiency

### **SQL Analysis Performed**

Used PostgreSQL to perform data analysis and generate business metrics from the Brazil e-commerce dataset.

### **Key SQL operations performed:**

- Used JOIN operations to combine customer, order, payment, and seller datasets
- Used GROUP BY for state-wise and seller-wise aggregations
- Used SUM() to calculate total revenue
- Used AVG() to calculate average order value and delivery time
- Used COUNT() to calculate total orders and order status distribution
- Used ORDER BY and LIMIT to identify top-performing sellers and states
- Used date-based aggregation for monthly revenue trend analysis
- Filtered and cleaned inconsistent or missing records before visualization

## **Power BI Dashboard Explanation**

The Power BI dashboard was designed to provide interactive business insights using visual analytics.

### **Dashboard components included:**

- **KPI Cards:**

- Average Delivery Days
- Total Revenue
- Average Order Value

- **Monthly Revenue Trend:**

Visualized revenue growth over time to identify seasonal sales patterns and business growth trends.

- **Revenue by State:**

Compared revenue contribution from different states and identified top-performing regions.

- **Top Sellers:**

Identified sellers generating the highest revenue.

- **Orders by Status:**

Analyzed operational efficiency by visualizing delivered, shipped, canceled, and processing orders.

- **Payment Distribution:**

Analyzed customer payment preferences including credit card, boleto, voucher, and debit card usage.

### **Key Insights Generated**

- São Paulo generated the highest overall revenue among all states.
- Revenue showed strong growth over time before stabilizing.
- Credit cards were the most commonly used payment method.
- Most customer orders were successfully delivered.
- Average delivery time was approximately 14 days.
- A small group of sellers contributed a large share of total revenue.